

Final report

Longitudinal Data Analysis

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1 Introduction

Our dataset is called "Chocolate cake", and as its name implies, it is about baking some batches of chocolate cake. The description given to us was the following:

*An experiment was conducted to determine the effect of baking temperature on quality for three recipes of chocolate cake. Recipes I and II differed in that the chocolate was added at 40°C and 60°C, respectively, while recipe III contained extra sugar. Six different baking temperatures were considered: 175°C, 185°C, 195°C, 205°C, 215°C, and 225°C. 45 batches of cake batter were prepared using each of the 3 recipes 15 times in a completely random order. Each batch was large enough for 6 cakes, and the six baking temperatures were randomly assigned to the 6 cakes per batch in a random manner. The humidity, if the mold was greased (**Greased_mold**), and the amount of sunflower oil and palm oil (**Sunfloweroil_amount**, **Palmoil_amount**) in the oven was registered. One of several measurements of quality made on each cake was the breaking angle of the cake. The goal is to characterize these **BreakingAngles**".*

This dataset has the characteristic of being a type of hierarchical model called "split plot model", which is further discussed in Section 6. For our analysis, however, we will not consider this structure, as it has not been worked in class and it turns out that it is not needed in order to study the breaking angles of interest.

2 Exploratory data analysis

To start this section we will describe our dataset, presenting each variable. We will then present an exploratory analysis in order to start to see some relationships between the variables and their mean and variance structure.

In Table 1 the type and the value and range of the variables are shown. Notice that in our dataset we have six categorical variables, whereas we only have two continuous numerical variables, one of them being our outcome, **BreakingAngles**. Note that variable **Repetition** does not have a particular meaning for our outcome, it is just a way to identify, jointly with the recipe, our batches of cake.

Variable	Type	Values
Recipe	Categorical	1, 2, 3
Repetition	Categorical	1 to 15
Temperature	Ordinal	175, 185, 200, 215, 225
Sunfloweroil_amount	Ordinal	65, 150, 165
Palmoil_amount	Binary	0, 75
Greased_mold	Binary	False, True
Humidity	Continuous	1.109 - 17.708
BreakingAngles	Continuous	18 - 63

Table 1: Description of the variables in our data set.

Next, we will focus on the average evolution of our data, which describes how the general profile of our data evolves over time. To do so, first we look at some summary statistics that might prove useful to analyse our data, which can be seen in Table 2.

Outcome	Temp.	n	min	max	mean	sd	median	q1	q3	iqr	se
BrAngles	175	45	19	47	28.0	7.05	26	24	32	8	1.05
BrAngles	185	45	18	46	30.0	7.26	28	25	33	8	1.08
BrAngles	195	45	21	51	31.4	7.50	30	27	35	8	1.12
BrAngles	205	45	19	49	32.2	7.35	30	27	37	10	1.10
BrAngles	215	45	21	57	35.8	8.39	35	31	40	9	1.25
BrAngles	225	45	20	63	35.4	9.03	35	30	37	7	1.35

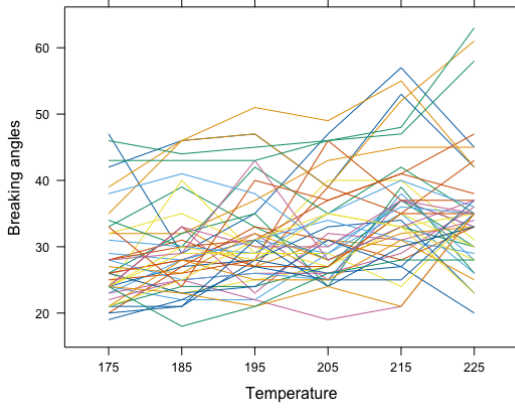
Table 2: Summary statistics for **BreakingAngles** according to **Temperature**.

From the above-presented table we can extract some important information which we will also plot in order to visualize it more easily.

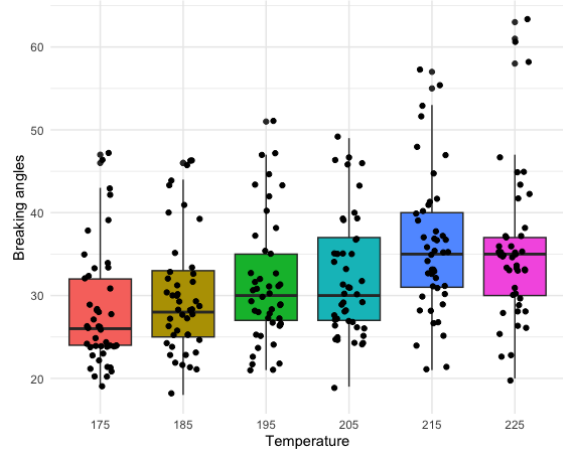
First of all, we can observe that there is an increment of the mean breaking angles of 7.4 degrees ($28.0 - 35.4$) from temperature 175 to temperature 225. Overall, the mean response increases with age, and so does the variance. There are a couple of temperatures in which this general tendency is not fulfilled, but we can affirm that the trend goes this way. This can also be seen in the two graphical representations shown in Figures 1(a) and 1(b). Notice that in Figure 1(a) there is a

linear tendency, which is a useful result in order to later fit a linear mixed model.

In Figure 1(b), it can be seen that the variance increases with temperature, and that there are several batches of cake that are quite far from the mean. The standard error also increases with the temperature, which makes sense, and can also be seen in Table 2.



(a) Spaghetti plot of the breaking angles as a function of the temperature.



(b) Boxplot of the breaking angles as a function of the temperature.

From the summary statistics presented in Table 2 we can also compute the increment per degree of temperature, resulting in

$$\Delta = \frac{\Delta \text{Breaking angles}}{\Delta \text{Temperature}} = 0.15.$$

So we get that for each increase of 1°C the breaking angle increases 0.15° . Note that we can use this type of summary since we have seen that the general tendency is a linear increase.

Now, let us look at the average evolution for some of the variables in our dataset, in order to see if there are differences among the different subgroups.

In Figure 1 we can see the plots for each variable according to the subgroups among them. For recipes, a spaghetti plot is shown, whereas for the rest of the variables, boxplots and scatter plots are presented as they show better the mean evolution.

It is hard to see any significant differences between them, as the average evolutions do not look that much different from one another. For variable **Recipe**, maybe we could say that there is a slight difference between Recipe 1 and the other two, so we could try adding it to our model and seeing if it gives a significant result.

We might also be interested in seeing differences among the different subjects, not just in the mean. In Figure 2 we can see how the breaking angles evolve with the temperature for each individual batch of cake. It can be seen that while the majority of them behave in a similar way, there are some individuals that have a higher intercept (and thus higher values) than the others. That is, there is something that might distinguish some individuals from the others.

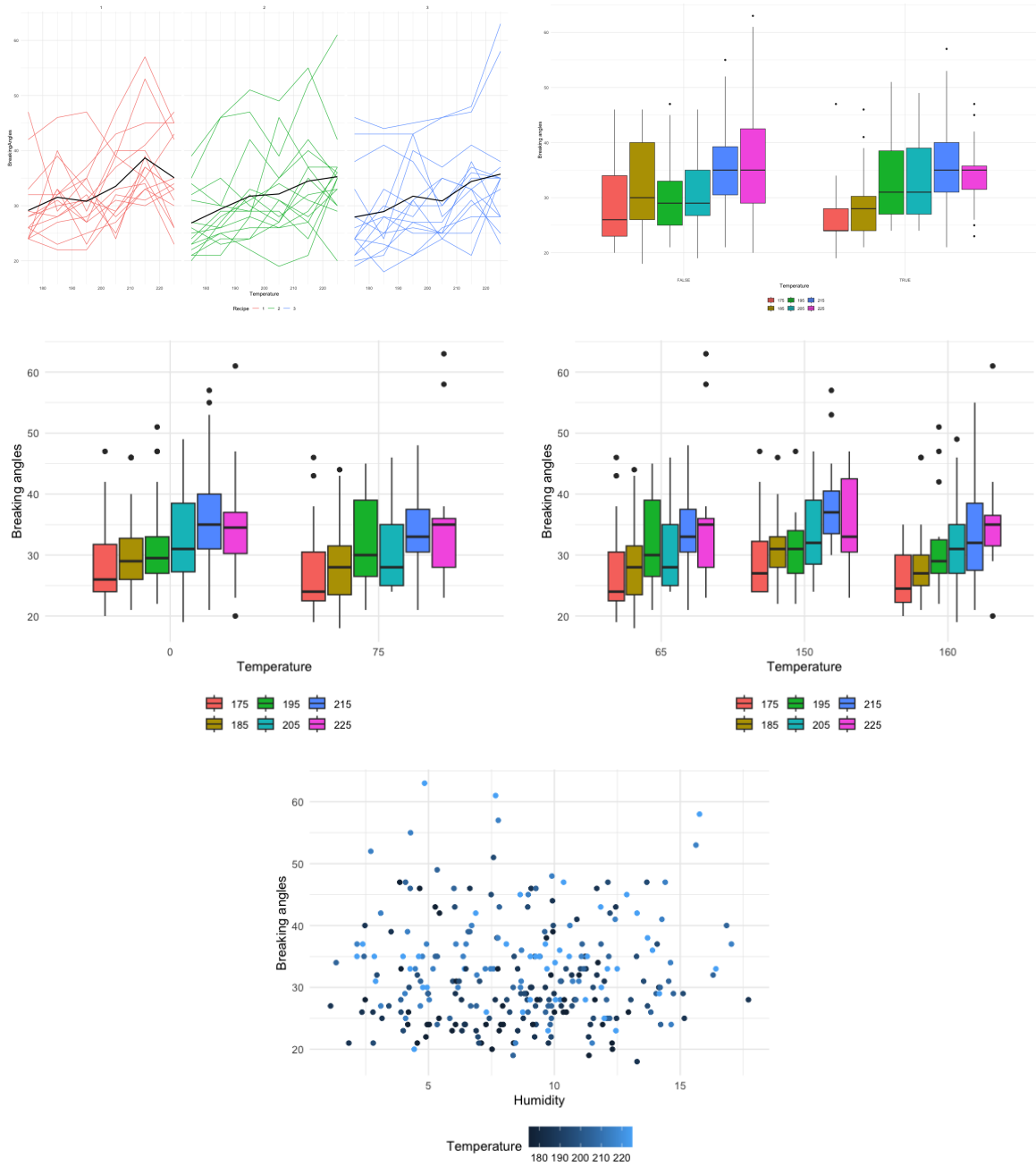


Figure 1: Mean evolution of the breaking angles across temperature for each subgroup of our variables.

Observing the figure, we can see that the individuals that look a bit more different than the others are the ones that correspond to Repetitions 1 & 2. This will be commented in the following sections.

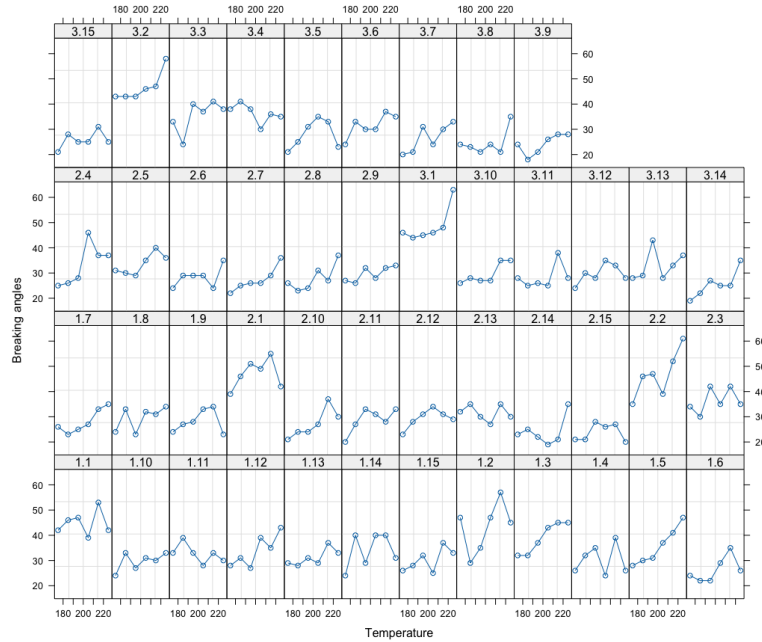


Figure 2: Breaking angles as a function of the temperature per individual batch of cake.

Now we can focus on the correlation structure. One characteristic of repeated data is that different measures are quite correlated among them, and we need to account for that in our model. In Figure 3 we can see the correlation between the different measures of the breaking angles at each temperature.

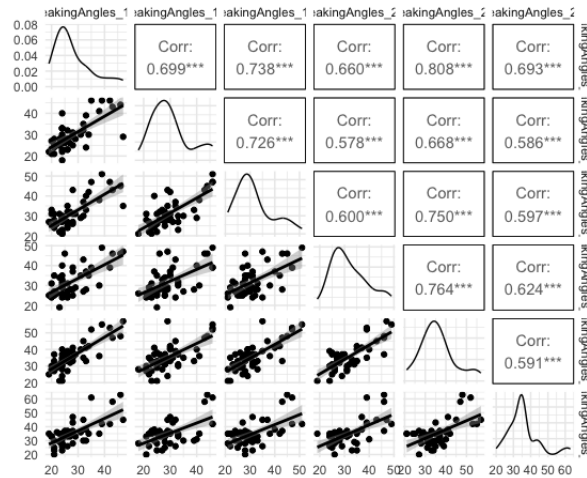


Figure 3: Correlation among measures.

A positive correlation is observed between all measurements, and it is high enough for all of them so that we need to take it into account. However, there is not a defined correlations structure, in the sense that we might not benefit from choosing a particular type of correlation. We will further explore the structure of this correlation matrix in Section 5.

2.1 Conclusions of the exploratory data analysis

To sum up the results we found in this first analysis, we have that:

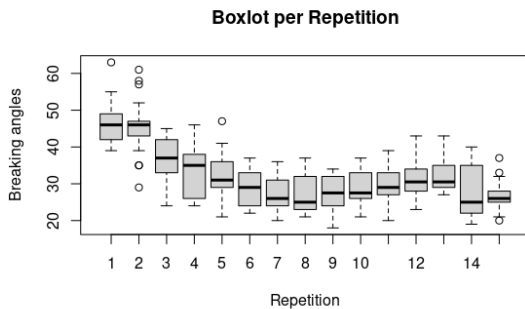
- Our dataset has a continuous outcome, six categorical variables and a continuous one.
- Overall, the mean and variance of the breaking angles increase linearly with the temperature.
- The increment in the mean breaking angles is of 7.4 from the initial temperature to the final one, with an increment per unit of temperature of 0.15 degrees.
- It is hard to see differences among the subgroups of all the variables. Maybe between recipes there is a slight difference.
- There is variability between individuals that we might want to account for in future models.
- There is a high correlation between the measurements, and the first impression is that the correlation does not follow a particular structure.

3 Linear model

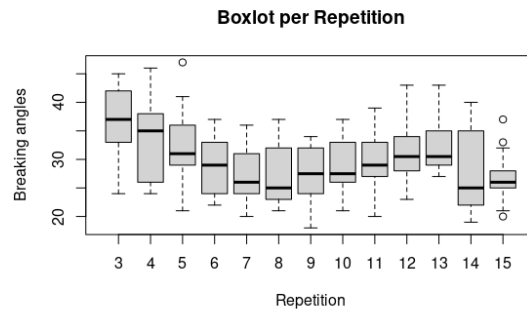
In this part, we will focus on observing the behaviour of a standard linear model. Following the equation $Y = AX + \epsilon$. Our goal here was to detect the importance of each variable and which one we can potentially eliminate.

3.1 Observing in terms of repetitions

As said before we didn't choose to work with the repetition as the time variable. But exploring a little bit we found that the first two repetition where a bit different from the other ones. Looking at the 4(b) we observe that the boxplot doesn't even touch.



(a) Boxplots of the repetition.



(b) Boxplot of the repetition without repetition 1 and 2

This can be a consequence of the way the measures were taken, see this link, since they were taken with a little manual kit. A slight change in the methodology of the measurement could have led to this result.

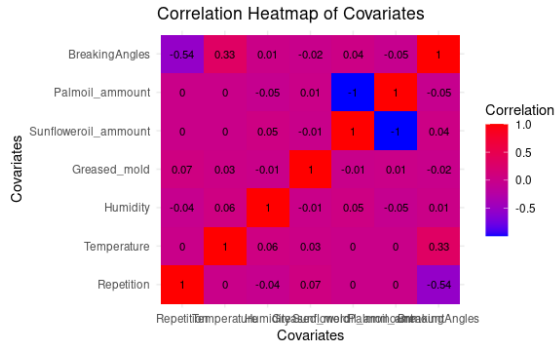
3.2 Variable selection

3.2.1 Correlation

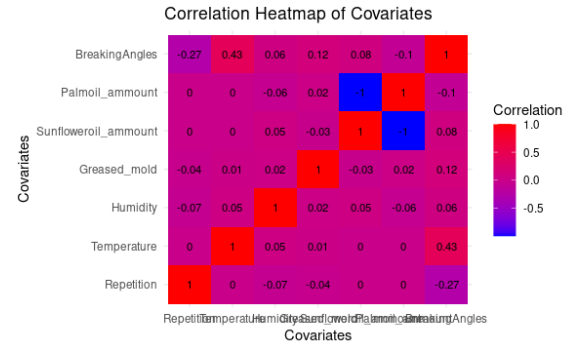
In order to select the most efficient ratio of covariates we first look at the correlation between the covariates. As explained before we can look at the correlation with and without the first 2 repetitions. The changes in the result is enormous in terms of correlation. For instance the correlation between `Repetition` and `BreakingAngles` is divided by two if we do not include the first two repetitions.

Notice that some of these correlations make no sense, as said in part 1, some of the covariates are categorical and this has not been taken into account here.

The principal information that we get from this correlation study is that the oil amounts are 100% correlated.



(c) Heatmap of the correlations with all the repetitions.



(d) Heatmap of the correlation ignoring repetition 1 and 2.

3.2.2 Stepwise selection

In order to arrive to a first conclusion that will help us build more complicated models, we choose to perform a stepwise selection to have a first view on the important variables. We also choose to use the standard AIC criteria in the stepwise selection (see in the Appendix the R code used).

We get the results :

Coefficient	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	15.95638	3.60996	4.420	1.47×10^{-5} ***
Recipe2	-1.41748	0.71233	-1.990	0.0477 *
Recipe3	-1.50715	0.71185	-2.117	0.0352 *
Repetition2	-1.20241	1.59204	-0.755	0.4508
Repetition3	-10.41647	1.60910	-6.473	5.00×10^{-10} ***
Repetition4	-13.70629	1.59311	-8.603	8.55×10^{-16} ***
Repetition5	-14.82128	1.60059	-9.260	$< 2 \times 10^{-16}$ ***
Repetition6	-18.33722	1.59490	-11.497	$< 2 \times 10^{-16}$ ***
Repetition7	-19.72610	1.59490	-12.368	$< 2 \times 10^{-16}$ ***
Repetition8	-19.82128	1.60059	-12.384	$< 2 \times 10^{-16}$ ***
Repetition9	-19.95221	1.60450	-12.435	$< 2 \times 10^{-16}$ ***
Repetition10	-18.45221	1.60450	-11.500	$< 2 \times 10^{-16}$ ***
Repetition11	-16.94444	1.59168	-10.646	$< 2 \times 10^{-16}$ ***
Repetition12	-16.41647	1.60910	-10.202	$< 2 \times 10^{-16}$ ***
Repetition13	-15.01981	1.59204	-9.434	$< 2 \times 10^{-16}$ ***
Repetition14	-19.24592	1.59739	-12.048	$< 2 \times 10^{-16}$ ***
Repetition15	-20.52369	1.59739	-12.848	$< 2 \times 10^{-16}$ ***
Temperature	0.15700	0.01702	9.223	$< 2 \times 10^{-16}$ ***
Greased_moldTRUE	1.35663	0.60713	2.234	0.0263 *

Notice that we do find here some high significances among the results. So this is a quite good first model, which we can use to build the next ones. In order to validate this model, we have studied the residuals produced. We can see on the following graphics, that the residuals seem to be independent and normally distributed. We can thus validate this first model.

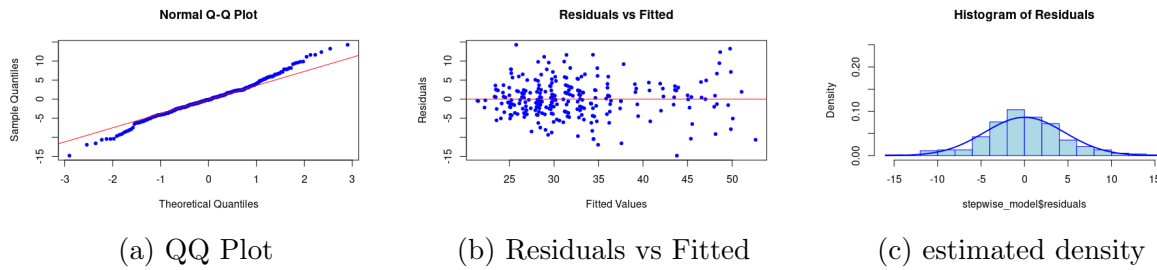


Table 3: Checking the residuals

3.3 Conclusion of the linear model part

Notice that, this conclusion is to be taken very carefully, knowing that this first model is not adapt to those data (as the other ones developed later are better). This conclusion is thus temporary and only here to help us going to a Linear Mixed Model (LMM).

From this first try of fitting a model, we can conclude that the most relevant variables are : **Repetition**, **Temperature**. And that the variables **Recipe**, **Greased_mold** are significant but less than the 2 before. And we can make the hypothesis that the variables **Sunfloweroil_amount**, **Palmoil_amount**, **Humidity** are not really needed.

Once we know how the covariates behave in the normal linear model, we can try to use a two stage analysis to get closer to the final objective of this work: to adjust a random-effects model.

4 Two stage analysis

In this section we will be using an explicit two stage analysis in order to get an initial impression on how the covariates behave. This is the preliminary step before fitting a linear mixed model, and will serve as a bridge to this model. In this section we have followed [2], where the two-stage analysis theoretical framework is greatly explained.

4.1 First stage

The goal in this first stage is to explain the within-subjects variability. With this objective in mind, we fit a linear regression model for each subject separately. Let us introduce some notation. Let Y_{ij} denote our response of interest (breaking angle), for the i th individual (batch of cake), measured at temperature (remember that temperature plays the role of time in our problem) T_{ij} , $i = 1, \dots, N$, $j = 1, \dots, n_i$, that is, $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{in_i})'$. In this first stage we assume that $\mathbf{Y}_i$ satisfies the linear regression model

$$Y_{ij} = \beta_{1i} + \beta_{2i}T_{ij} + \beta_{3i}H_{ij} + \epsilon_{ij}, \quad j = 1, \dots, n_i. \quad (1)$$

Where $\beta_{1i}, \beta_{2i}, \beta_{3i}$ are the unknown subject-specific regression coefficients. T_{ij} and H_{ij} are, respectively, the temperature and the humidity, in the j th measurement for the i th batch of cake. Finally, ϵ_{ij} is the error term. Despite the fact that exploratory analysis and linear model points that humidity is not an informative covariate, we have decided to add it into the first stage with the aim to assess its effect and significance.

Let us analyze the results of the $N = 45$ fitted linear models. In the Figure 4, the reader can see two illustrative plots explaining the results of the model. In the left, there are the confidence interval for the estimates. As the reader can see, none of the three (intercept, temperature and humidity) show significance on the estimates, in general. Only few estimations seem to be significant, since they do not include the 0 in the confidence interval. Moreover, in the right graphic on Figure 4, we can see the relationship between the estimates of the intercept and the slope (understanding the slope as the coefficient accompanying the temperature). The reader can observe that there is a negative relation between these coefficients. In addition, we have computed the correlation, which is -0.93 . It is a high negative correlation, as we can expect observing Figure 4. Besides the correlation between β_{1i}, β_{2i} , we have computed the correlation between β_{1i}, β_{3i} and β_{2i}, β_{3i} . These correlations are -0.023 and -0.26 , respectively. Meaning that the correlation is still negative, but not as high as in the β_{1i}, β_{2i} case.

4.2 Second stage

The main objective in the second stage analysis is to explain the variability between subjects. To do this, we present the multivariate regression model

$$\begin{cases} \beta_{1i} = \beta_1 R_{1i} + \beta_2 R_{2i} + \beta_3 R_{3i} + \sum_{l=1}^{15} \beta_{l+3} Repl_i + b_{1i} \\ \beta_{2i} = \beta_{19} R_{1i} + \beta_{20} R_{2i} + \beta_{21} R_{3i} + \sum_{l=1}^{15} \beta_{l+21} Repl_i + b_{2i} \\ \beta_{3i} = \beta_{39} R_{1i} + \beta_{40} R_{2i} + \beta_{41} R_{3i} + \sum_{l=1}^{15} \beta_{l+41} Repl_i + b_{3i} \end{cases} \quad (2)$$

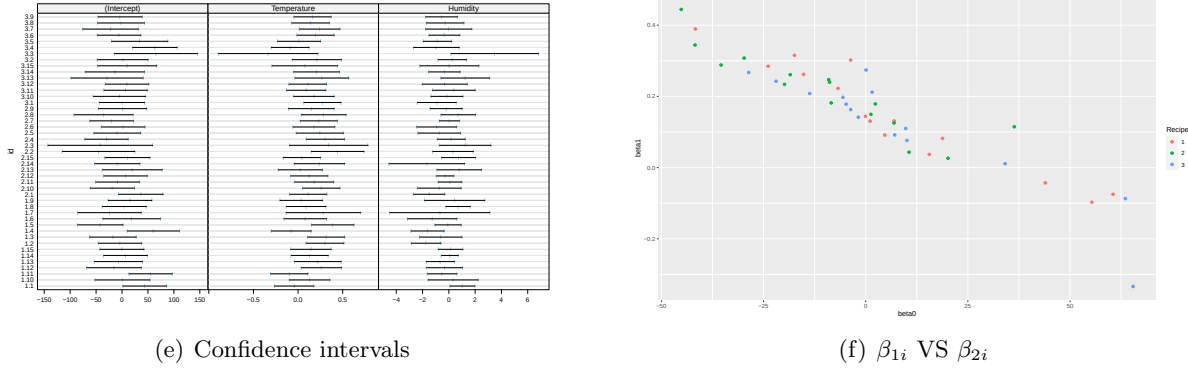


Figure 4: In the left, the confidence intervals of the estimates in (1). In the right, β_{1i} VS β_{2i} .

Where R_{1i}, R_{2i}, R_{3i} are indicator variables defined to be one if the batch i is from Recipe 1, 2 or 3 respectively. And $Rep_{1i}, \dots, Rep_{15i}$ are also indicator variables which are one if batch i belongs to the repetition 1, $\dots$ or 15. By taking the multivariate regression model in (2), the subject-specific intercepts, temperature and humidity effects are related to the recipe of the batch (either recipe one, two or three), and with the repetition of the batch (repetition one to fifteen). Here it is important to comment that other multivariate regression models have been fitted. We have tried to use the sunflower oil amount, and palm oil amount as a grouping variables, due the fact that all the batches in Recipe 1, 2 and 3 have the same sunflower oil amount. And, in the case of the palm oil amount, in the Recipe 1 and 2 we have the same amount, and in recipe 3 another different. Finally, we have not included these variables because its poor significance.

The parameters $\beta_1, \beta_2, \beta_3$ represents the average response at the beginning (first temperature used) for each recipe. And the same for β_4 to β_{18} , this time for each repetition. The parameters β_{19}, β_{20} and β_{21} represents the average temperature effects for each recipe separately. Moreover, the parameters β_{22} to β_{38} represents the average temperature effects for each repetition separately. Analogously we can interpret the meaning of the remaining parameters, these will be related with humidity.

In Table 7 the reader can observe three tables where the summaries of the models fitted are exposed. The first observation is that, in general, the estimates obtained are not significant. In the p -values column of each of three subtables in Table 7, we can see that only few estimations are significant at a 0.05 level. Notwithstanding, it is interesting to interpret at least the sign of the estimates obtained. In the case of the linear model fitted with β_{1i} (see Table 4) as a response variable, we can see that all the repetitions except the number 4 have a negative estimate. Meaning that for those batches coming from repetitions 2 to 15, except 4, the average response at the beginning is smaller than in repetition 1 (which is the one acting as a baseline). Moreover, in Table 6 we can see that the estimates for recipe 2 and 3 are positive. Meaning that the average humidity effects for those batches coming from recipe 2 and 3 tends to be bigger than the ones for batches from recipe 1. Finally, we can see that all the estimates for the parameter β_{2i} (see Table 5), which is the one related with the temperature, are all near to 0, and non-significant. This means that the recipe and the repetition do not vary the temperature coefficients, which indicates that these covariates are not affecting the effects of the temperature on the braking angles.

Coefficient	Estimate	Std Error	T Value	P Value
(Intercept)	31.78	16.56	1.92	0.07
Recipe2	-15.82	9.84	-1.61	0.12
Recipe3	0.89	9.84	0.09	0.93
Repetition2	-42.55	22.00	-1.93	0.06
Repetition3	-24.71	22.00	-1.12	0.27
Repetition4	4.65	22.00	0.21	0.83
Repetition5	-32.34	22.00	-1.47	0.15
Repetition6	-21.61	22.00	-0.98	0.33
Repetition7	-48.69	22.00	-2.21	0.04
Repetition8	-37.64	22.00	-1.71	0.10
Repetition9	-22.41	22.00	-1.02	0.32
Repetition10	-34.19	22.00	-1.55	0.13
Repetition11	-8.79	22.00	-0.40	0.69
Repetition12	-26.30	22.00	-1.20	0.24
Repetition13	-31.89	22.00	-1.45	0.16
Repetition14	-32.00	22.00	-1.45	0.16
Repetition15	-19.92	22.00	-0.91	0.37

Table 4: Summary of the linear model fitted for the first stage β_{1i} .

Coefficient	Estimate	Std Error	T Value	P Value
(Intercept)	0.10	0.09	1.12	0.27
Recipe2	0.07	0.05	1.24	0.23
Recipe3	-0.03	0.05	-0.52	0.61
Repetition2	0.20	0.12	1.68	0.10
Repetition3	-0.01	0.12	-0.06	0.96
Repetition4	-0.07	0.12	-0.55	0.59
Repetition5	0.10	0.12	0.83	0.42
Repetition6	0.04	0.12	0.31	0.76
Repetition7	0.14	0.12	1.14	0.26
Repetition8	0.06	0.12	0.48	0.63
Repetition9	0.00	0.12	0.01	0.99
Repetition10	0.07	0.12	0.61	0.54
Repetition11	-0.06	0.12	-0.47	0.65
Repetition12	0.05	0.12	0.41	0.68
Repetition13	0.06	0.12	0.47	0.64
Repetition14	0.08	0.12	0.64	0.53
Repetition15	-0.03	0.12	-0.23	0.82

Table 5: Summary of the linear model fitted for the first stage β_{2i} .

Coefficient	Estimate	Std Error	T Value	P Value
(Intercept)	-0.64	0.55	-1.16	0.25
Recipe2	0.19	0.33	0.58	0.57
Recipe3	0.37	0.33	1.12	0.27
Repetition2	0.08	0.73	0.11	0.91
Repetition3	1.83	0.73	2.50	0.02
Repetition4	-0.33	0.73	-0.45	0.65
Repetition5	-0.09	0.73	-0.13	0.90
Repetition6	-0.38	0.73	-0.51	0.61
Repetition7	0.24	0.73	0.33	0.74
Repetition8	0.84	0.73	1.15	0.26
Repetition9	0.36	0.73	0.49	0.63
Repetition10	0.28	0.73	0.38	0.71
Repetition11	0.45	0.73	0.61	0.55
Repetition12	0.16	0.73	0.21	0.83
Repetition13	0.93	0.73	1.27	0.21
Repetition14	-0.18	0.73	-0.25	0.81
Repetition15	0.76	0.73	1.04	0.31

Table 6: Summary of the linear model fitted for the first stage β_{3i} .

Table 7: Summary of the multivariate regression model in (2).

4.3 Combined model and conclusions

To finish this section, one can write the combined model by substituting the equations on (2) in (1), obtaining

$$\begin{aligned}
Y_{ij} = & (\beta_1 R_{1i} + \beta_2 R_{2i} + \beta_3 R_{3i} + \sum_{l=1}^{15} \beta_{l+3} \text{Repl}_i + b_{1i}) + (\beta_{19} R_{1i} + \beta_{20} R_{2i} + \beta_{21} R_{3i} \\
& + \sum_{l=1}^{15} \beta_{l+21} \text{Repl}_i + b_{2i}) T_{ij} + (\beta_{39} R_{1i} + \beta_{40} R_{2i} + \beta_{41} R_{3i} + \sum_{l=1}^{15} \beta_{l+41} \text{Repl}_i + b_{3i}) H_{ij} + \epsilon_{ij}.
\end{aligned} \tag{3}$$

In conclusion, we have seen that adding **Recipe** and **Repetition** do not make significant changes on the coefficients of the intercept, temperature and humidity. Moreover on the first stage we have seen that none these are significant. Then, as we were suspecting in the previous sections we are dealing with a hard problem. Nonetheless, this two-stage analysis has given us an idea on how to deal with the linear mixed model that we will fit in the next section. We have seen that **Recipe** and **Repetition** are not informative variables, in addition humidity is not explaining well the breaking angle. So we arrive to the last part of the work knowing that the only useful variable, and the one that we will focus on when fitting the linear mixed model, is the temperature.

5 Linear mixed model

In this final section of the work we will formulate a plausible random-effects model. With this purpose, we use all the knowledge about the problem acquired in the previous sections. We have seen that there are few significant and informative variables in the data. Thanks to the exploratory analysis, linear model and two-stage analysis we know with which covariates to work in order to obtain the better random-effects model.

Our intuition, and the analysis provided above, points to a simple random-effects model, accounting for the Repetition covariate significance (see on the linear model section) and for the Temperature as well. Notwithstanding, we have followed the natural path to assess that our intuition is right, that is, to start by fitting the simplest possible model and adding different components while checking which is the best model.

5.1 Fitting several random-effects models

We have fitted a model with a random intercept, that is (using the notation introduced in the previous section)

$$Y_{ij} = (\beta_0 + b_{0i}) + \beta_1 T_{ij} + \epsilon_{ij}, \quad (4)$$

where $b_0 \sim N(0, d_{00})$ and $\epsilon \sim N(0, \sigma^2)$. It is important to remark that these b_0 's are now random effects, while in the two-stage analysis they were not (thus, we are committing an abuse of notation here). Notice that this random intercept accounts for the variability explained by the Repetition covariate. Remember that, by the construction of our problem, the individuals (or batches) are defined by Recipe and Repetition. Then, by taking a random intercept for each batch, what we are doing is explaining the variability caused by each batch (and then for the recipe and repetition) in the intercept. Moreover, we have seen in the exploratory analysis and in the linear model section, that the repetition is an informative and significant variable in our problem, while the Recipe is not that important. The results in terms of the estimates of the fixed effects and the variance of the random effects are shown in Table 8.

Random effects:			
Groups	Name	Variance	Std.Dev.
id	Intercept	40.25	6.34
Residual		20.71	4.55
Fixed effects:			
	Estimate	Std. Error	t value
(Intercept)	0.516	3.389	0.152
Temperature	0.158	0.016	9.746
Correlation of Fixed Effects:			
	(Intr)		
Temperature	-0.957		

Table 8: Summary of the random and fixed effects of the model (4).

In addition, exploiting the fact that they are nested models, we have compared via the likelihood ratio test the normal linear model with only the Temperature as a covariate. We have obtained $\lambda_{LR} = 170$, which is well-known to be chi-squared distributed by the Wilks' Theorem. So, in conclusion, we reject the null hypothesis that the random effect is 0, and then we prefer the random intercept model. Moreover, we have computed the Akaike's Information Criterion (AIC) and Bayesian Information Criterion (BIC) for both models. $AIC = 1877.304$, $BIC = 1888.099$ in the normal linear model case, and $AIC = 1709.298$, $BIC = 1723.692$ in the random intercept model. Thus, both criterion supports that the random-effect model is better.

Now, we fit a random slope model, it is

$$Y_{ij} = \beta_0 + (\beta_1 + b_{1i})T_{ij} + \epsilon_{ij}, \quad (5)$$

where $b_{1i} \sim N(0, d_{11})$. In this case we have obtained the results shown in Table 9. As the reader can observe, the value for the variance is very near to 0. For this reason R returns a **Warning** message about the non-proper convergence of the model. This low variance is an indication that in the slope there is not a clear effect on the batch of cake.

Random effects:			
Groups	Name	Variance	Std.Dev.
id	Temperature	0.001	0.032
Residual		20.45	4.52
Fixed effects:			
	Estimate	Std. Error	t value
(Intercept)	0.516	3.235	0.159
Temperature	0.158	0.0168	9.409
Correlation of Fixed Effects:			
	(Intr)		
Temperature	-0.956		

Table 9: Summary of the random and fixed effects of the model (5).

For the random slope model we have also used the likelihood-ratio test, and we have concluded that the random slope model is better than the linear model (with only the temperature as a covariate). Notwithstanding, as commented before, the low variance of the random effect suggests that it is not necessary to use this random effect.

Despite the fact that we have seen that the random slope is not necessary, let us fit a random-effects model with a random intercept and a random slope and check the results. The model is the following,

$$Y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})T_{ij} + \epsilon_{ij}, \quad (6)$$

where $(b_0, b_1) \sim N(0, D)$ and $\epsilon_i \sim N(0, \Sigma_i)$. When fitting the previous model in R we have convergence errors. Using either restricted maximum likelihood (REML) criterion or maximum likelihood,

the convergence is not reached and then we cannot fit (6) in practice. However, by taking simpler variance-covariance matrix of the random effects than the unstructured one, we are able to fit the model. We have fitted (6) taking

$$D_d = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix} \quad \text{or} \quad D_{id} = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_1^2 \end{pmatrix}.$$

We have used ANOVA to compare these models. The AIC and BIC obtained for both models are: $AIC = 1706.794$, $BIC = 1721.158$ in the case of D_{id} , and $AIC = 1708.751$, $BIC = 1726.706$ in the D_d case. Hence, since both AIC and BIC are smaller, the model with D_{id} is preferable. Moreover, we see that the AIC is slightly better than in the random intercept model, but the BIC is worse. Remember that BIC penalizes more the model complexity than AIC. The estimates of the fixed effects of this model are exposed in Table 10.

	Value	Std Error	DF	t-value	p-value
(Intercept)	0.5158730	3.235279	224	0.159452	0.8735
Temperature	0.1580317	0.016796	224	9.408638	0.0000

Table 10: Summary of the random intercept and slope model with D_{id} matrix of variance covariance of the random effects.

Nonetheless, the model with a random intercept and a random slope will not be the chosen one. As commented above, the random slope is not of our interest. Moreover, the model above, taking D_{id} , is forcing the two random effects (intercept and slope) to have the same variance. And, as has been discussed before and exposed in Table 8 and Table 9, the variance of the random intercept is much more bigger than the variance of the random slope. Thus, it does not make sense to force these variances to be equal.

5.2 Applying some transformations

At this point, after checking different random effects model, and taking into account all that has been seen in the previous sections of the work, we choose the random intercept model (4). This is the most plausible random-effects model. Now, our goal is to check if there is any transformation that can improve the performance of the model.

We try to take the logarithm in the response variable, obtaining the following model,

$$\log(Y_{ij}) = (\beta_0 + b_{0i}) + \beta_1 T_{ij} + \epsilon_{ij}, \quad (7)$$

where $b_0 \sim N(0, d_{00})$ and $\epsilon \sim N(0, \sigma^2)$. The results of this random-effects model are presented in the Table 11. As the reader can observe the value of the variance estimated for the random effect is much more lower than in the random intercept model without transformation previously fitted. While the estimate in this case is 0.0335 (see Table 11), the estimate in the random intercept model is 40.25 (see Table 8).

Moreover, we have computed the AIC and the BIC for both, the transformed model and the model without the logarithm. $AIC = 1709.298$, $BIC = 1723.692$ for the model without transformation

Random effects:			
Groups	Name	Variance	Std.Dev.
id	Intercept	0.0335	0.183
Residual		0.0194	0.139
Fixed effects:			
	Estimate	Std. Error	t value
(Intercept)	2.4503602	0.1035	23.734
Temperature	0.0049	0.0005	9.967
Correlation of Fixed Effects:			
	(Intr)		
Temperature	-0.961		

Table 11: Summary of the random and fixed effects of the model (7).

(4), and $AIC = -164.5144$, $BIC = -150.121$ for the model taking the logarithm in the response (7). It is, the AIC and BIC for the transformed model are much more smaller, meaning that the latter model is preferable than the one fitted at the beginning of the section.

5.3 Further analysis of the model

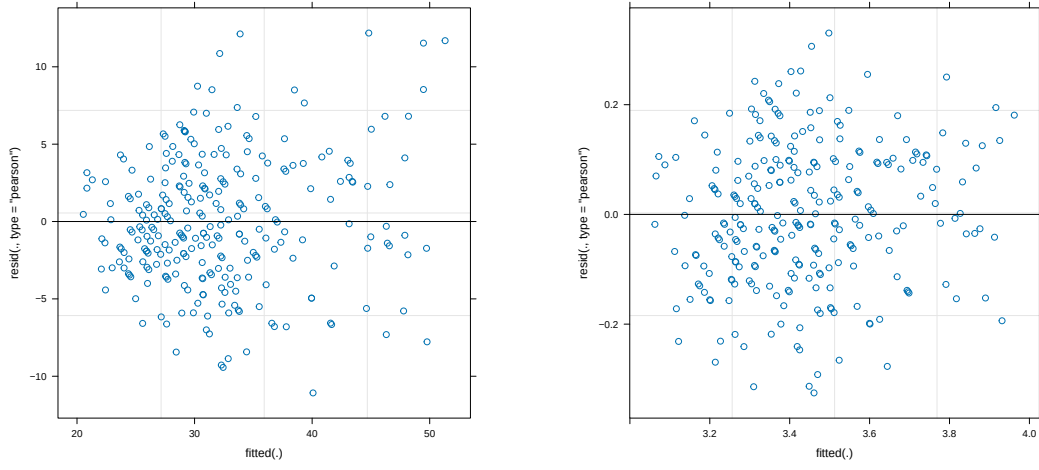
Another option is to play with the correlation structure of the residuals. There are several structures of correlation matrix that can be used. We have tried some of them: compound symmetry, autoregressive and general correlation matrix with no structure. In all these cases the fitted model does not improve the results in terms of AIC and BIC. Hence, we chose the random-effects model with the transformation, but with no correlation structure of the residuals explicitly specified.

Moreover, we have wondered whether the usage of weights for model the variance gives good results and the performance of the model is improved. Notwithstanding, there are no clear differences between the standardized residuals against the fitted values when splitting in the different categorical variables we have. Thus, the usage of weights is discarded.

5.4 Residuals and conclusions

Finally we should take a look into the residuals of the model. The Pearson's residuals against the fitted values are exposed in the Figure 5. In the left part the reader can see the residuals against the fitted values in the case of the random intercept model (not transformed) (4), and in the right the residuals against the fitted values for the random intercept model taking the logarithm for the response variable (7). As we can observe, the residuals in the right graphic seem to be more symmetric with respect the black line than in the left graphic. This supports our election of the model with the logarithmic transformation.

In conclusion, after all the analysis exposed in this section, our ultimate choice of a random-effects model is the one with a random intercept and with the response variable transformed via the logarithm (7). This is the model with better AIC and BIC. Moreover, as we have been commenting during all section, this is the most suitable model. A random slope has been discarded and other covariates are not significant, as we saw in previous sections. The random intercept is explaining



(a) Residuals of the random-effects model (4)

(b) Residuals of the random-effects model (7)

Figure 5: In the left, Pearson's residuals against the fitted values for the random effect model (4). In the right, the same but for model (7).

the variability due to the different repetitions. These, and another reasons pointed out during the section, make us choose (7) as the most plausible random-effects model.

6 Split plots model

In this section we will expand our analysis, since there is a kind of model named split plots model. We will first explain this model formally and justify why it seems suitable to our data and, then, apply it in RStudio and interpret the results.

A split-plot design is characterized by separate random assignments of levels of factors, where levels of some factor are assigned to larger experimental units called whole plots. Each whole plot is divided into smaller units, called split-plots, and levels of another factor are randomly assigned to split-plots [3]. In the Figure 6 we can see a visual explanation of this definition.

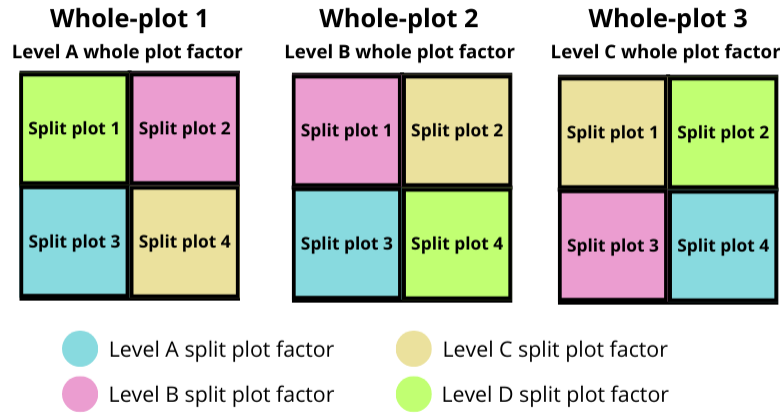


Figure 6: Split Plot design

In our case the whole-plots will be the 15 different batches, each batch is assigned randomly a recipe, with the objective of having 3 batches per recipe. And we divide this whole-plot into the 6 cakes each batch has, each cake would be a split plot, and we randomize one baking temperature to each cake (we have 6 cakes and 6 baking temperatures). We can see the structure of our problem in Table 12.

Batch 1		Batch 2		...	Batch 15	
Recipe 1		Recipe 1			Recipe 3	
Cake 1 T ^o 1	Cake 2 T ^o 2	Cake 1 T ^o 1	Cake 2 T ^o 2		Cke 1 T ^o 1	Cake 2 T ^o 2
Cake 3 T ^o 3	Cake 4 T ^o 4	Cake 3 T ^o 3	Cake 4 T ^o 4		Cake 3 T ^o 3	Cake 4 T ^o 4
Cake 5 T ^o 5	Cake 6 T ^o 6	Cake 5 T ^o 5	Cake 6 T ^o 6		Cake 5 T ^o 5	Cake 6 T ^o 6

Table 12: Structure of our problem

Finally, we present the general formula for this model. As we have two different sizes of experimental units, we also need two error terms to model the corresponding experimental errors. We need one error term "acting" on the plot level and another one on the subplot level [1]. Let Y_{ijk} be the observed response of the k the replicate of a plot with level of the whole plot factor i and level j of the split plot factor. We use the model

$$Y_{ijk} = \mu + \alpha_i + \eta_{k(i)} + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk},$$

Where α_i is the fixed effect of the whole plot factor, β_j is the fixed effect of split plot factor and $(\alpha\beta)_{ij}$ is the corresponding interaction term. We call $\eta_{k(i)}$ the whole-plot error. It is expressed with a nesting notation, this means that η_k depends on i that is the whole plot factor level. At the end we have ϵ_{ijk} which is the split-plot error. We use the assumptions

$$\eta_{k(i)} \text{ i.i.d. } \sim N(0, \sigma_\eta^2), \quad \epsilon_{ijk} \text{ i.i.d. } \sim N(0, \sigma^2)$$

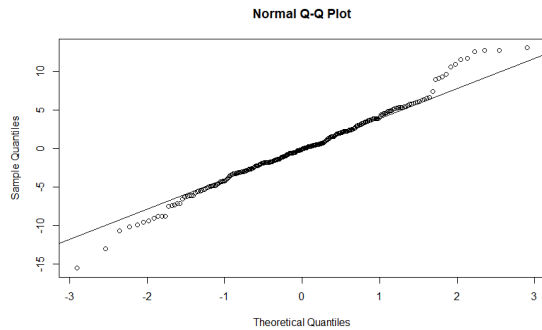
In our case the equation will be:

$$Y_{ijk} = \mu + \alpha_i + \eta_{k(i)} + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk},$$

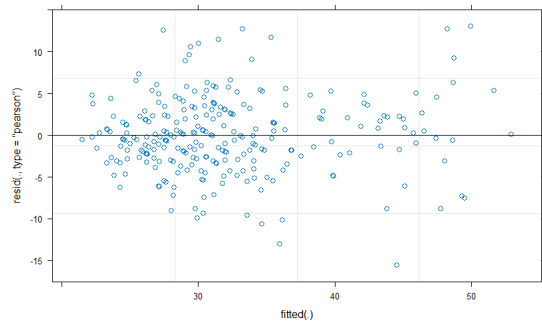
where :

- α_i is the fixed effect of the recipe $i \in \{1, 2, 3\}$.
- β_j is the fixed effect of temperature $j \in \{1, \dots, 6\}$.
- $(\alpha\beta)_{ij}$ is the interaction term between recipe i and temperature j .
- $\eta_{k(i)}$ the whole-plot error.
- ϵ_{ijk} which is the split-plot error.

This model can be fitted in RStudio via the function `lmer` from the library `lmerTest`. But before seeing the results of our model, we need to check the heterocedasticity and normality of the residuals in order to accept the results as valid. We will do that with Figures 7(a) & 7(b), the first graph shows that the residuals approximate heterogeneity, and the second graph confirms normality, hence we can use the model.



(a) Normality



(b) Homocedasticity

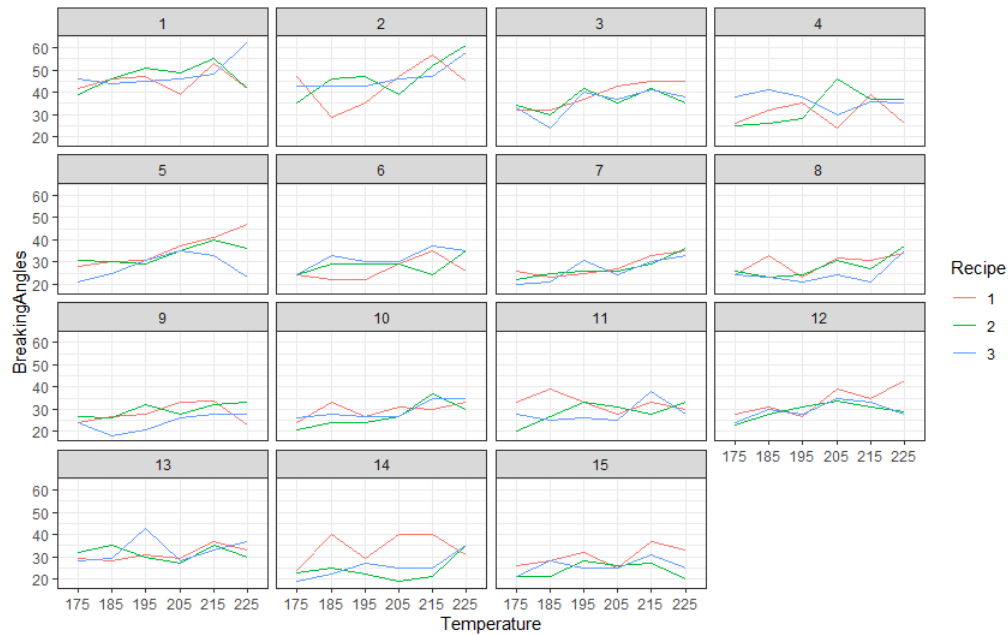


Figure 7: Spaghetti plot for each batch

First, we can see the result for the significance of each factor with the function `anova` applied at the model which returns

```

Type III Analysis of Variance Table with Satterthwaite's method
      Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
Recipe      135.09   67.54      2    238  2.9242  0.05564 .
Temperature 2100.30  420.06      5    238 18.1859 2.821e-15 ***
Recipe:Temperature 205.98   20.60     10    238  0.8918  0.54156
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

We can see that at level 0.05, the fixed effect of temperature is significant. But neither the recipe by itself or the interaction between recipe and temperature are significant for the breaking angle of the cake.

A confirmation to this significance can be found in Figure 7 where we plotted each batch as a block, the Breaking angles registered in each temperature for each recipe. We can see that there is no difference between the recipes in any batch, but that the differences in each temperature (the slope of the spaghetti plot) are changing a lot in each measurement, meaning that the temperature does change the Breaking angle of the cake.

We can also plot the predicted response for each temperature and recipe by our model, we can see them in Figure 8 where, we confirm again that the recipe does not affect the response, but that the temperatures do. In the aforementioned figure the breaking angle is increasing as the temperature does, with the maximum in 215°.

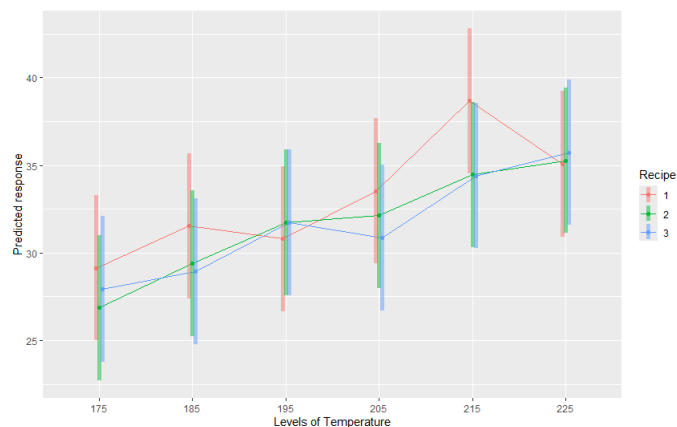


Figure 8: Predicted response per recipe and temperature

7 Comment on missing data

This project had the characteristic that some of the data sets could present missing data. In our data, however, it is not the case. If we had had some missing data, it would have been important to take it into account, especially if the missings were in the outcome variable, that is, in the breaking angles.

Missing data is important because it might lead to biased estimates, such as attrition bias, it can lower the statistical power and it might make us inflate the variability in the data. To mitigate the impact of missing data in longitudinal analysis, strategies such as multiple imputation, maximum likelihood estimation, and sensitivity analyses are used. These approaches aim to account for missing data patterns and produce more robust and reliable results.

References

- [1] Lukas Meier. *ANOVA and mixed models: a short introduction using R*. Chapman and Hall/CRC, 2022.
- [2] Geert Verbeke, Geert Molenberghs, and Geert Verbeke. *Linear mixed models for longitudinal data*. Springer, 1997.
- [3] Hao Zang. *Split-Plot designs*. 2020. URL: https://www.stat.purdue.edu/~zhanghao/STAT514/Lecture_Notes/LectureNotes20-Split-plot-Designs-.html.

A Annex: R code

Following the reader can see the code used in this work. There is more code and material in our GitHub repositories <https://github.com/adinbartoli/LDA> and https://github.com/ArnauGF/MSc-UPC/tree/longitudinal_data_analysis.

A.1 Exploratory data analysis

```
library(readxl)
library(dplyr)
library(lattice)
library(ggplot2)

# Exploratory data analysis

choccake <- read_xlsx("choccake.xlsx")
choccake <- choccake %>%
  mutate(id = paste0(choccake$Recipe, ".", choccake$Repetition),
         Sunfloweroil_amount = Sunfloweroil_ammount, Palmoil_amount = Palmoil_ammount) %>%
  mutate_at(vars(id, Recipe, Repetition, Greased_mold, Sunfloweroil_amount, Temperature,
                 Palmoil_amount), as.factor) %>%
  select(-Sunfloweroil_ammount, - Palmoil_ammount)

plot0 <- xyplot(BreakingAngles ~ Temperature, group = id,
               data = choccake,
               type = "l", xlab = "Temperature", ylab = "Breaking angles")
plot0

# Exploring the mean and variance

library(labelled) # labeling data
library(rstatix)  # summary statistics
library(ggpubr)   # convenient summary statistics and plots
library(GGally)   # advanced plot
library(car)      # useful for anova/wald test
library(Epi)      # easy getting CI for model coef/pred
library(lme4)     # linear mixed-effects models
library(lmerTest) # test for linear mixed-effects models
library(emmeans)  # marginal means
library(multcomp) # CI for linear combinations of model coef
library(geepack)  # generalized estimating equations
library(ggeffects) # marginal effects, adjusted predictions
library(gt)       # nice tables

library(tidyverse) # for everything (data manipulation, visualization, coding, and more)
theme_set(theme_minimal() + theme(legend.position = "bottom")) # theme for ggplot

group_by(choccake, Temperature) %>%
  get_summary_stats(BreakingAngles)

choccake$Temperature <- as.numeric(choccake$Temperature)
```

```
# First we look at it generally

ggplot(choccake, aes(x = Temperature, y= BreakingAngles, fill = Temperature)) +
  geom_boxplot() +
  geom_jitter(width = 0.2) +
  guides(fill = "none") +
  labs(x = "Temperature", y = "Breaking angles")

# Now we can look at it for some variables

ggplot(choccake, aes(x = Greased_mold, BreakingAngles, fill = Temperature)) +
  geom_boxplot() +
  labs(x = "Temperature", y = "Breaking angles", fill = "")

mean <- choccake %>%
  group_by(Recipe, Temperature) %>%
  summarise(BreakingAngles = mean(BreakingAngles))
mean

ggplot(choccake, aes(Temperature, BreakingAngles, col = Recipe)) +
  geom_line(aes(group = factor(id))) +
  geom_line(data = mean, col = "black", lwd = 1) +
  facet_wrap(~ Recipe)

ggplot(choccake, aes(x = Humidity, BreakingAngles, col = Temperature)) +
  geom_point() +
  labs(x = "Humidity", y = "Breaking angles", fill = "")

ggplot(choccake, aes(Temperature, BreakingAngles, col = factor(Sunfloweroil_amount))) +
  geom_line(aes(group = factor(id))) +
  geom_line(data = mean, col = "black", lwd = 1) +
  facet_wrap(~ Sunfloweroil_amount)

ggplot(choccake, aes(x = Sunfloweroil_amount, BreakingAngles, fill = Temperature)) +
  geom_boxplot() +
  labs(x = "Temperature", y = "Breaking angles", fill = "")

ggplot(choccake, aes(Temperature, BreakingAngles, col = factor(Palmoil_amount))) +
  geom_line(aes(group = factor(id))) +
  geom_line(data = mean, col = "black", lwd = 1) +
  facet_wrap(~ Palmoil_amount)

ggplot(choccake, aes(x = Palmoil_amount, BreakingAngles, fill = Temperature)) +
  geom_boxplot() +
  labs(x = "Temperature", y = "Breaking angles", fill = "")

# Looking at the data per individual

ggplot(choccake, aes(Temperature, BreakingAngles, col = factor(id))) +
  geom_point() +
```

```

geom_line() +
facet_wrap(~ id) +
labs(x = "Temperature", y = "Breaking angless", col = "") +
guides(col = guide_legend(nrow = 3))

choccake2 <- groupedData(BreakingAngles ~ Temperature|id,
                        choccake,
                        outer = ~id,
                        labels = list(y = "Breaking angles"))

plot2 <- plot(choccake2)
plot2

# Let's look at the correlation now

ggpairs(select(choccakeWide, starts_with("Breaking")),
lower = list(continuous = "smooth"))

```

A.2 Linear model

```

initial_model <- lm(BreakingAngles ~ ., data = data)
stepwise_model <- step(initial_model, direction = "both", criterion = "AIC")
summary(stepwise_model)

```

A.3 Two stage analysis

```

modlist2<-lmList(BreakingAngles~Temperature + Humidity,
                data=choccake2, na.action=na.pass)
plot(intervals(modlist2))
ts2 <- data.frame(Id=as.factor(attributes(modlist2)$names),
                 beta0=coef(modlist2)[,1],beta1=coef(modlist2)[,2],
                 beta2=coef(modlist2)[,3])

char_var <- as.character(ts2$Id)
split_var <- strsplit(char_var, "\\.")
Recipe.ts2 <- sapply(split_var, '[', 1)
Repetition.ts2 <- sapply(split_var, '[', 2)
Recipe.ts2 <- as.numeric(Recipe.ts2)
Repetition.ts2 <- as.numeric(Repetition.ts2)
#finally we add recipe and repetition to the ts2 dataframe
ts2 <- cbind(ts2, Recipe = as.factor(Recipe.ts2),
            Repetition = as.factor(Repetition.ts2))

plot2 <- ggplot(ts2)+
  aes(x=beta0, y=beta1, color=Recipe)+
  geom_point()
with(ts2, cor(beta0,beta1))
with(ts2, cor(beta0,beta2))

```

```

with(ts2, cor(beta1,beta2))
modbeta0.2<-lm(beta0~Recipe + Repetition,data=ts2)
summary_model<-summary(modbeta0.2)
modbeta1.2<-lm(beta1~Recipe + Repetition, data = ts2)
summary_model <- summary(modbeta1.2)
modbeta2.2<-lm(beta2~Recipe + Repetition, data = ts2)
summary_model<-summary(modbeta2.2)

```

A.4 Linear mixed model

```

library(nlme)
library(lme4)
model.lm<- lm(BreakingAngles ~ Temperature,data=choccake)
summary(model.lm)
model.inter<- lmer(BreakingAngles ~ Temperature + (1 | id),data=choccake)
summary(model.inter)
value1<-as.numeric(2*(logLik(model.inter)-(logLik(model.lm))))
value1
p1<- 0.5*(1-pchisq(value1,1))
p1
AIC(model.inter)
BIC(model.inter)
AIC(model.lm)
BIC(model.lm)
model.slope<- lmer(BreakingAngles ~ Temperature + (Temperature-1 | id),
                  data=choccake)
summary(model.slope)
value2<-as.numeric(2*(logLik(model.slope)-(logLik(model.lm))))
value2
p2<- 0.5*(1-pchisq(value2,1))
p2
AIC(model.slope)
BIC(model.slope)
AIC(model.lm)
BIC(model.lm)
mod.diag<-lme(BreakingAngles~Temperature,random=list(id=pdDiag(~Temperature)),
              data=choccake)
summary(mod.diag)
mod.id<-lme(BreakingAngles~Temperature,random=list(id=pdIdent(~Temperature)),data=choccake)
summary(mod.id)
anova(mod.id, mod.diag)
anova(model.inter, mod.id)
AIC(model.inter)
BIC(model.inter)
model.inter.transf <- lmer(log(BreakingAngles) ~ Temperature + (1 | id),data=choccake)
summary(model.inter.transf)
anova(model.inter, model.inter.transf)
AIC(model.inter)
BIC(model.inter)

```

```

AIC(model.inter.transf)
BIC(model.inter.transf)
plot(model.inter)
plot(model.inter.transf)
mod <- lme(log(BreakingAngles)~Temperature, random=~1|id, data = choccake)
summary(mod)
mod.corr <- lme(log(BreakingAngles)~Temperature, random=~1|id, data = choccake,
               correlation=corCAR1(form=~1|id))
summary(mod.corr)
mod.corr1 <- lme(log(BreakingAngles)~Temperature, random=~1|id, data = choccake,
               correlation=corCompSymm(form=~1|id))
summary(mod.corr1)
mod.corr2 <- lme(log(BreakingAngles)~Temperature, random=~1|id, data = choccake,
               correlation=corAR1(form=~1|id))
summary(mod.corr2)
mod.corr2 <- lme(log(BreakingAngles)~Temperature, random=~1|id, data = choccake,
               correlation=corCAR1(form=~1|id))
summary(mod.corr2)
mod.corr3 <- lme(log(BreakingAngles)~Temperature, random=~1|id, data = choccake,
               correlation=corSymm(form=~1|id))
summary(mod.corr3)
mod.wei <- lme(log(BreakingAngles)~Temperature, random=~1|id, data = choccake,
               weights = varIdent(form = ~1|Repetition))
summary(mod.wei)
plot(mod, resid(.,type="pe")~fitted(.)|Repetition)
plot(mod.wei, resid(.,type="pe")~fitted(.)|Repetition)
plot(mod, resid(.,type="pe")~fitted(.)|Greased_mold)

```

A.5 Split plots model

```

# Model fit
library(lmerTest)
fit.cake <- lmer(BreakingAngles ~ Recipe * Temperature + (1 | Repetition),
               data = data)

# Homocedasticity and Normality plots
plot(fit.cake)
qqnorm(resid(fit.cake))
qqline(resid(fit.cake))

# Spaghetti plot by recipe
library(ggplot2)
ggplot(aes(x = Temperature, y = BreakingAngles, group = Recipe, colour = Recipe), data = data) + geom_line()
  facet_wrap(~ Repetition) + theme_bw()

# Predicted response for each temperature and recipe by our model

library(emmeans)
emmip(fit.cake,Recipe~Temperature,CIs = TRUE,type = "response")

```

